B.7 External Interface

B.7.1 Introduction

While the computational aspects of *Mathematica* are essentially identical on all kinds of computers that run *Mathematica*, the external interface to *Mathematica* inevitably varies somewhat from one computer system to another.

Mathematica is usually divided into two parts: the *kernel* that actually does computations, and a *front end* that deals with interaction with the user. The kernel is set up to be as similar as possible on every different kind of computer; the front end varies from one computer to another, taking advantage of particular features of each computer.

The kernel and front end of *Mathematica* communicate by exchanging packets of standard printable ASCII. (The front end can also usually send interrupt signals to the kernel.) The kernel generates all graphics in PostScript form.

Under multi-tasking operating systems (such as UNIX), the front end is usually a separate process, which communicates with the kernel through a bidirectional pipe.

Sometimes the front end may be a generic program, such as a shell window or an Emacs editor; in other cases, the front end may be a special program built specially for use with *Mathematica*. The same goes for PostScript interpreters: on some systems *Mathematica* may use built-in PostScript interpreters; on others, a special PostScript interpreter program may be needed.

The front end and kernel need not both be on the same computer; they can running on separate computers, connected by a network or other communications link.

This Section primarily concerns the external interface to the kernel of *Mathematica*. It does not discuss in any detail issues associated with the front end.

Even in discussing the external interface to the kernel, there are many details that depend on the operating system you are using. What is said in this Section mostly applies to UNIX-like operating systems. It should nevertheless be valid, at least in general terms, for any operating system that supports analogs of pipes and shell files.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; http://www.wolfram.com/; 1-800-441-6284.

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